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Thermodynamic description of the Al–Sb–Y system

L.G. Zhang[a], H.M. Chen[a], H.Q. Dong[a], G.X. Huang[a], L.B. Liu[a,b,*], Z.P. Jin[a]

[a] School of Material Science and Engineering, Central South University, Changsha, Hunan 410083, PR China
[b] Key Laboratory of Non-ferrous Materials Science and Engineering, Ministry of Education, Central South University, Changsha 410083, PR China

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ABSTRACT

The phase equilibria of the Sb–Y and Al–Sb–Y systems were analyzed and a complete thermodynamic description of this ternary system was obtained. The thermodynamic descriptions of the other two binary systems were taken from the literature. Most of the binary intermetallic phases, except Sb_3Y_5, were assumed to have no ternary solubility. Based on the experimental data, the calculated phase equilibria were in good agreement with the experimental data.

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1. Introduction

The interest in aluminum alloys was continuously increasing, especially because of their application in the automotive and aerospace industries. Micro-alloying had been used to improve the high-temperature strength of selected age hardenable Al alloys throughout the years [1–3]. Rare earth elements in conventional aluminum alloys had shown beneficial effects to the casting process [4–6]. These elements reduced the contents of gases and some impurities and the spacing between secondary dendrite arms. The use of rare earth Y as a micro-alloying element in Al had been studied for several years. In addition to the beneficial effect to the casting process, it was reported that the mechanic properties including peak hardness, over-aging behavior and the tensile strength were improved with Y added in aluminum alloys [7]. Sb was also an important addition to improve the microstructure and mechanical properties of cast aluminum alloys [8–9].

The knowledge of the thermodynamics and phase diagram was one of the basic requirements in the field of materials research and process control of new alloy systems. The present work aimed to evaluate the Al–Sb–Y ternary system and to develop a thermodynamic description of the system by means of the CALculation of PHAse Diagram (CALPHAD) technique [10].

* Corresponding author at: School of Material Science and Engineering, Central South University, Changsha, Hunan 410083, PR China. Tel.: +86 7318877732; fax: +86 7318876692.
E-mail address: pdc@mail.csu.edu.cn (L.B. Liu).

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2. Experiment information

2.1. Al–Sb and Al–Y binary system

The Al–Sb system was thermodynamically studied by Coughanowr et al. [11], as shown in Fig. 1.

The Al–Y system was firstly studied by Lukas [12]. Recently, Liu et al. [19] reassessed this binary system by thermodynamic calculation based on new experimental data. In this work, the thermodynamic parameters by Liu et al. [13] were adopted, as shown in Fig. 2.

2.2. Sb–Y binary system

Used thermal analysis, X-ray diffractography, and metallographic methods, Schmidt and McMasters [14] had established the phase diagram of Sb–Y system. Four intermetallic compounds, having the formulae SbY, Sb_3Y_4, Sb_3Y_5 and SbY_3, were identified in this system. The structures of these phases were summarized in Table 1. The SbY phase melted congruently at 2310 °C while SbY_3 and Sb_3Y_5 phases result from peritectic reactions at 1240 °C and 1690 °C. And the Sb_3Y_4 phase formed peritectically at 2120 °C but decomposed by a eutectoid reaction at 1660 °C. Two eutectic reactions ($L \leftrightarrow$ Hcp-Y + SbY_3 and $L \leftrightarrow$ SbY + Rom-Sb) were observed by Schnidit and McMasters [14], as well as three peritectic reactions ($L + SbY \leftrightarrow Sb_3Y_4$, $L + Sb_3Y_4 \leftrightarrow Sb_3Y_5$, and $L + Sb_3Y_5 \leftrightarrow SbY_3$), and two eutectoid reactions ($Sb_3Y_4 \leftrightarrow Sb_3Y_5 + SbY$ and Bcc_Y $\leftrightarrow$ L + Hcp_Y).

Based on the results of Schnidit and McMasters [14], Massalski [15] constructed the phase diagram of Sb–Y binary system.

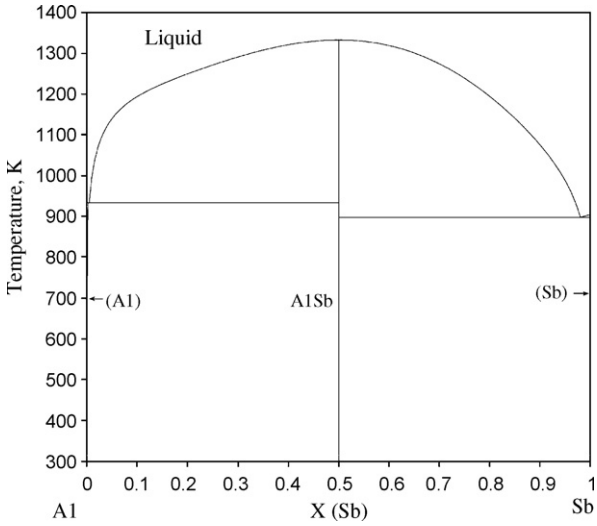

Fig. 1. The calculated Al–Sb phase diagram by Coughanowr et al. [11].

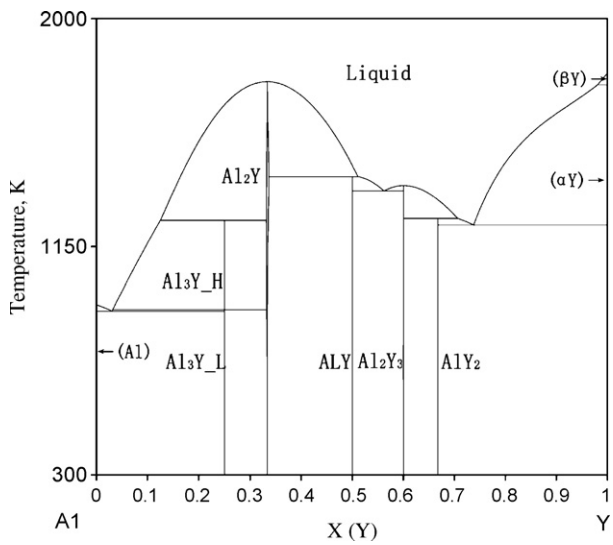

Fig. 2. The calculated Al–Y phase diagram by Liu et al. [13].

The heats of formation of the Sb–Y alloys at 300 K had been measured using a differential direct isoperibol calorimetry by Borsese et al. [16].

2.3. Al–Sb–Y ternary system

The Al–Sb–Y ternary system was first investigated by Murav'eva et al. [17]. The partial isothermal section within the limits of 0–33.3 at.% Y and 0–50.0 at.% Sb at 773 K was reported. In their work, the tie lines of Al–SbY, AlSb–SbY and Al_3Y_L–SbY were found. And they also reported two existed three-phase regions (Al + AlSb + SbY and Al + SbY + Al_3Y_L). No ternary compound was found in their work.

Zeng and Wang [18] investigated the 800 K isothermal section of Al–Sb–Y system by XRD, DTA, SEM and optical microscopy methods. It was reported that there were 12 single-phase regions, 23 two-phase regions, and 10 three-phase regions. They also investigated three sub-binary systems to identify the binary compounds. In their work, most of the binary compounds were confirmed except the AlY_2 phase. With the X-ray powder diffraction, they suggested a AlY_3 phase substituted for AlY_2 phase in the Y-rich region. The maximum solid solubility of Al in Sb_3Y_5 was reported to be about 27 at.%.

Recently, in order to confirm the Y-rich phase is AlY_2 or AlY_3, Liu et al. [19] reinvestigated the whole Al–Y phase diagram. They found the Al-rich phase was AlY_2. In this work, AlY_2 phase was considered to be stable phase in this region.

Up to now, there is no experimental or calculated thermochemistry of the Al–Sb–Y system available in the literature. Therefore, the Al–Sb–Y system has been optimized on basis of the ternary phase diagram information.

Table 1
Crystallographic data of intermetallic phases.

Phase	Lattice parameters			Structure type, space group
	a (nm)	*b* (nm)	*c* (nm)	
AlSb	0.6135			SZn, $F\bar{4}3m$
Al_3Y_H	0.61884		2.1094	$BaPb_3$, $R\bar{3}m$
Al_3Y_L	0.6276		0.4582	Ni_3Sn, $P63/mmc$
Al_2Y	0.78611			$MgCu_2$, $Fd\bar{3}m$
AlY	0.3884	1.1522	0.4385	BCr, $Cmcm$
Al_2Y_3	0.82411		0.76411	Al_2Zr_3, $P42/mnm$
AlY_2	0.654	0.508	0.940	Co_2Si, $Pnma$
SbY	0.6165			ClNa, $Fm\bar{3}m$
Sb_3Y_5	0.89114		0.62960	Mn_5Si_3, $P63/mmc$
SbY_3	1.2361		0.6180	PTi_3, $P42/n$

3. Thermodynamic model

3.1. Solution phases

The lattice stabilities for element Al, Sb and Y are referred to Dinsdale [20], an ordinary substitutional solution model is employed to describe liquid, Fcc, Bcc, Hcp and Rhom terminal solution. The mole Gibbs energy of a solution phase Φ (Φ = liquid, Fcc, Bcc, Hcp and Rhom) can be represented as a sum of the weighted Gibbs energy for the pure components, the ideal entropy term describing a random mixing of the components, and the excess Gibbs energy describing the degree of deviation from ideal mixing, i.e.:

$$G^{\phi} = \sum_{i=\mathrm{Al,Sb,Y}} x_i {}^0G_i^{\phi} + RT \sum_{i=\mathrm{Al,Sb,Y}} x_i \ln(x_i) + {}^{\mathrm{ex}}G^{\phi} \quad (1)$$

where

$$ {}^{\mathrm{ex}}G^{\phi} = x_{\mathrm{Al}}x_{\mathrm{Sb}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Sb}})^{j(j)} L^{\phi}_{\mathrm{Al,Sb}} + x_{\mathrm{Al}}x_{\mathrm{Y}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Al}} - x_{\mathrm{Y}})^{j(j)} L^{\phi}_{\mathrm{Al,Y}} + x_{\mathrm{Sb}}x_{\mathrm{Y}} \sum_{j=0,1\ldots}^{N} (x_{\mathrm{Sb}} - x_{\mathrm{Y}})^{j(j)} L^{\phi}_{\mathrm{Sb,Y}} + x_{\mathrm{Al}}x_{\mathrm{Sb}}x_{\mathrm{Y}} L^{\phi}_{\mathrm{Al,Sb,Y}} \quad (2)$$

where ϕ denotes the solution phases, x_i (i = Al, Sb and Y) denotes mole fraction of component i, and ${}^0G_i^{\phi}$ is the molar Gibbs energy of pure element i in the structural state of ϕ. ${}^{(j)}L^{\phi}_{\mathrm{Al,Sb}}$ and ${}^{(j)}L^{\phi}_{\mathrm{Al,Y}}$ are taken from Ref. [11,13]. And ${}^{(j)}L^{\phi}_{\mathrm{Sb,Y}}$ is describing as

$$ {}^{(j)}L^{\phi}_{\mathrm{Sb,Y}} = A_j + B_j T \quad (3)$$

A_j and B_j are the adjusted parameters being optimized in the present work.

$L^{\phi}_{\mathrm{Al,Sb,Y}}$ is ternary interaction parameters. Due to lack of experiment data, it is set to be zero.

3.2. Binary intermetallic phases

Most of the binary intermetallic phases are treated as stoichiometric phases, which are described as M_pN_q. According to Neumann–Kopp rule, the Gibbs energy of the phase M_pN_q is formulated as

$$G_{M_pN_q} = \frac{p}{p+q}{}^0G_M^{\mathrm{HSER}} + \frac{q}{p+q}{}^0G_N^{\mathrm{HSER}} + C + DT \quad (4)$$

where M and N denote Al, Sb and Y, C and D are the adjusted parameters being optimized in the present work.

According to Zeng and Wang [18], the Sb_3Y_5 phase has the homogeneity region in the Al–Sb–Y ternary system. The Gibbs energy expression of this phase is formalized as

$$G^{(\mathrm{Al,Sb})_{0.375}\mathrm{Y}_{0.625}} = Y_{\mathrm{Al}}^I G_{\mathrm{Al:Y}} + Y_{\mathrm{Sb}}^I G_{\mathrm{Sb:Y}} + 0.375RT(Y_{\mathrm{Al}}^I \ln Y_{\mathrm{Al}}^I + Y_{\mathrm{Sb}}^I \ln Y_{\mathrm{Sb}}^I) + Y_{\mathrm{Al}}^I Y_{\mathrm{Sb}}^I L_{\mathrm{Al,Sb:Y}} \quad (5)$$

where $G_{\mathrm{Al:Y}}$, $G_{\mathrm{Sb:Y}}$ and $L_{\mathrm{Al,Sb:Y}}$ are expressed as

$$G_{\mathrm{Al:Y}} = 0.375{}^0G_{\mathrm{Al}}^{\mathrm{Fcc}} + 0.625{}^0G_{\mathrm{Y}}^{\mathrm{Hcp}} + E_1 + F_1T \quad (6)$$

$$G_{\mathrm{Sb:Y}} = 0.375{}^0G_{\mathrm{Sb}}^{\mathrm{Fcc}} + 0.625{}^0G_{\mathrm{Y}}^{\mathrm{Hcp}} + E_2 + F_2T \quad (7)$$

$$L_{\mathrm{Al,Sb:Y}} = E_3 + F_3T \quad (8)$$

The $G_{\mathrm{Al:Y}}$ represents the Gibbs energy of the assumed Al_3Y_5 compound with Sb_3Y_5 structure.

E and F are the adjusted parameters being optimized in the present work.

4. Results and discussion

4.1. The Sb–Y binary system

On the basis of lattice stabilities cited from Dinsdale [20], the optimization of the Sb–Y system was carried out using the PanOptimizer of Pandat software [21]. PanOptimizer is a software package for evaluating the thermodynamic, kinetic and thermo-physical model parameters from experimental or theoretically calculated data by solving the weighted least squares problem. More detailed information about PanOptimizer can be found in Ref. [22]. The phase diagram and thermochemical data were used as input to the program for the optimization. Each piece of selected information was given a certain weight by personal judgment, and verified by trial and error method during the assessment, until most of the selected experimental information was reproduced. All the evaluated parameters were listed in Table 2.

The optimal set of parameters was obtained as the sum of squares steadily approached a reasonably small value. To verify, we first checked the absolute residuals of each experimental data to ensure them within or around the uncertainties. Secondly, based on the list of values of each parameter (in the past 100 iterations) provided by PanOptimizer, we manually calculated the relative standard derivation (R.S.D.) and make sure they are small in the end of optimization. In a recent beta version of Pandat 8.0, the R.S.D.s of all parameters during optimization are automatically calculated. We used this version to double-check the reported parameters in Table 2. It confirms the validity of our assessment.

Fig. 3 shows the calculated phase diagram of Sb–Y system compared with experimental data, while Table 3 lists the invariant reactions in the Sb–Y system. An agreement within 10 K between the calculated and experimentally determined temperatures for these invariant reactions has been obtained. As shown in Fig. 4, the calculated heats of formation at 300 K are reasonably agreement with the

Table 2
Thermodynamic parameters of Al–Sb–Y system.

Phase	Thermodynamic parameters	Ref.
Liquid: (AL, Sb, Y)	${}^0L_{\mathrm{Al:Y}} = -192571.2 + 29.03622T$	[13]
	${}^1L_{\mathrm{Al:Y}} = -46742.9 + 0.46159T$	[13]
	${}^2L_{\mathrm{Al:Y}} = -70259.0 + 29.99999T$	[13]
	${}^0L_{\mathrm{Al:Sb}} = -41894.5 + 185.62130T - 22.67728T\ln(T)$	[12]
	${}^1L_{\mathrm{Al:Sb}} = -5685.8 + 1.93333T$	[12]
	${}^2L_{\mathrm{Al:Sb}} = -17158.5 - 9.54781T$	[12]
	${}^0L_{\mathrm{Sb:Y}} = -283141.0 + 10.64460T$	This work
	${}^1L_{\mathrm{Sb:Y}} = -33814.3 + 28.55170T$	This work
	${}^2L_{\mathrm{Sb:Y}} = -83240.7$	This work
Bcc_Y	${}^0L_{\mathrm{Sb,Y}}^{\mathrm{Bcc}} = -191265.1$	This work
Hcp_Y	${}^0L_{\mathrm{Sb,Y}}^{\mathrm{Hcp}} = -181277.2$	This work
AlSb: $(\mathrm{Al})_{0.5}(\mathrm{Sb})_{0.5}$	$G_{\mathrm{Al:Sb}}^{\mathrm{AlSb}} = -24486 + 7.528437T - 0.827827T\ln(T) + 0.0026797T^2 + 0.5GHSERAL + 0.5GHSERSB$	[12]
Al3Y_H: $(\mathrm{Al})_{0.75}(\mathrm{Y})_{0.25}$	$G_{\mathrm{Al:Y}}^{\mathrm{Al_3Y_H}} = -49236.3 + 5.10000T + 0.75GHSERAL + 0.25GHSERY$	[13]
Al3Y_L: $(\mathrm{Al})_{0.75}(\mathrm{Y})_{0.25}$	$G_{\mathrm{Al:Y}}^{\mathrm{Al_3Y_L}} = -49411.1 + 5.29096T + 0.75GHSERAL + 0.25GHSERY$	[13]
Al_2Y: $(\mathrm{Al,Y})_{0.667}(\mathrm{AL,Y})_{0.333}$	$G_{\mathrm{Al:Al}}^{\mathrm{Al_2Y}} = 5000 + GHSERAL$	[13]
	$G_{\mathrm{Y:Y}}^{\mathrm{Al_2Y}} = 5000 + GHSERY$	[13]
	$G_{\mathrm{Al:Y}}^{\mathrm{Al_2Y}} = -66528.4 + 9.42976T + 0.667GHSERAL + 0.333GHSERY$	[13]
	$G_{\mathrm{Y:Al}}^{\mathrm{Al_2Y}} = +66528.4 - 9.42976T + 0.333GHSERAL + 0.667GHSERY$	[13]
	${}^0L_{\mathrm{Al,Y:Y}} = {}^0L_{\mathrm{Al,Y:Al}} = {}^0L_{\mathrm{Al:Al,Y}} = {}^0L_{\mathrm{Y:Al,Y}} + 6667.0$	[13]
AlY: $(\mathrm{Al})_{0.5}(\mathrm{Y})_{0.5}$	$G_{\mathrm{Al:Y}}^{\mathrm{AlY}} = -54911.9 + 5.48155T + 0.5GHSERAL + 0.5GHSERY$	[13]
Al_2Y_3: $(\mathrm{Al})_{0.4}(\mathrm{Y})_{0.6}$	$G_{\mathrm{Al:Y}}^{\mathrm{Al_2Y_3}} = -48406.3 + 4.82944T + 0.4GHSERAL + 0.6GHSERY$	[13]
AlY_2: $(\mathrm{Al})_{0.333}(\mathrm{Y})_{0.667}$	$G_{\mathrm{Al:Y}}^{\mathrm{AlY_2}} = -44886.6 + 7.39566T + 0.333GHSERAL + 0.667GHSERY$	[13]
SbY: $(\mathrm{Sb})_{0.5}(\mathrm{Y})_{0.5}$	$G_{\mathrm{Sb:Y}}^{\mathrm{SbY}} = -117626.0 + 6.37709T + 0.5GHSERSB + 0.5GHSERY$	This work
Sb_3Y_4: $(\mathrm{Sb})_{0.429}(\mathrm{Y})_{0.571}$	$G_{\mathrm{Sb:Y}}^{\mathrm{Sb_3Y_4}} = -86145.4 - 3.90726T + 0.429GHSERSB + 0.571GHSERY$	This work
Sb_3Y_5: $(\mathrm{Al,Sb})_{0.375}(\mathrm{Y})_{0.625}$	$G_{\mathrm{Al:Y}}^{\mathrm{Sb_3Y_5}} = -43499.8 + 2.99898T + 0.375GHSERAL + 0.625GHSERY$	This work
	$G_{\mathrm{Sb:Y}}^{\mathrm{Sb_3Y_5}} = -91535.8 + 3.34727T + 0.375GHSERSB + 0.625GHSERY$	This work
	${}^0L_{\mathrm{Al,Sb:Y}} = -5000.0$	This work
SbY_3: $(\mathrm{Sb})_{0.25}(\mathrm{Y})_{0.75}$	$G_{\mathrm{Sb:Y}}^{\mathrm{SbY_3}} = -64919.6 + 4.62831T + 0.25GHSERSB + 0.75GHSERY$	This work

Note: Gibbs energies are expressed in J/mol. Lattice stabilities of elements Al, Sb and Y are referred to Dinsdale [20].

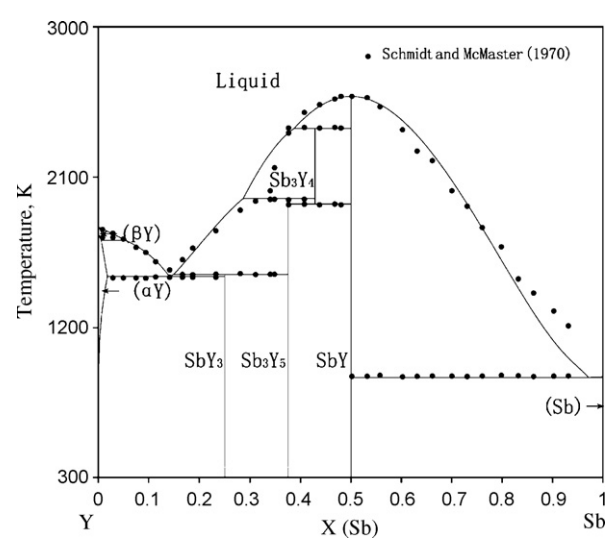

Fig. 3. The calculated Sb–Y phase diagram with the experimental data [14].

Table 3
Invariant reactions in the Sb–Y system.

Reaction	Composition of liquid (at.% Sb)	T/K	Type	Reference
Bcc_Y ⇔ L + Hcp_Y	–	1735	Eutectoid	[14]
	5.3	1730	Eutectoid	This work
L ⇔ Hcp_Y + SbY_3	14.5	1493	Eutectic	[14]
	16.4	1499	Eutectic	This work
L + Sb_3Y_5 ⇔ SbY_3	–	1513	Peritectic	[14]
	14.8	1515	Peritectic	This work
L + Sb_3Y_4 ⇔ Sb_3Y_5	–	1963	Peritectic	[14]
	28.6	1970	Peritectic	This work
Sb_3Y_4 ⇔ Sb_3Y_5 + SbY	–	1933	Eutectoid	[14]
	–	1937	Eutectoid	This work
L + SbY ⇔ Sb_3Y_4	–	2393	Peritectic	[14]
	38.6	2389	Peritectic	This work
L ⇔ SbY	–	2583	Congruent	[14]
	–	2582	Congruent	This work
L ⇔ SbY + Rhom_Sb	–	902	Eutectic	[14]
	97.1	897	Eutectic	This work

experiment data. It is demonstrated that the experimental data of thermodynamically properties can be well described by the present thermodynamic parameters within the experimental errors.

4.2. The Al–Sb–Y ternary system

There were no thermochemical and liquid projection information, so the ternary interaction parameters of liquid were considered as zero. Based on the phase relations of Zeng and Wang [18], the Sb_3Y_5 phase was optimized. The parameters were listed in Table 2.

The calculated Al–Sb–Y isothermal section at 800 K with Pandat software is shown in Fig. 5. The comparison of the calculated results and experimental results is listed in Table 4. Except the intermetallic compound AlY_3, which have not taken into consideration because of contradicting with the Al–Y binary system, the calculated isothermal section have well reproduced the experimental data. From our calculated results, we can find that the homogeneity region of Sb_3Y_5 extend up to 26.9 at.% Al, which are good agreement with the work of Zeng and Wang [18] (27 at.% Al).

Additionally, the isothermal section at 773 K is calculated, illustrated in Figs. 6. And this result can also well reproduce the work of Murav'eva et al. [17]. The liquids projection is further calculated

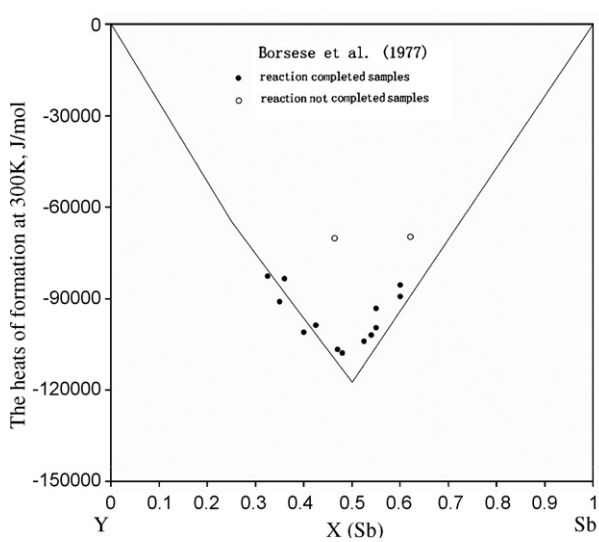

Fig. 4. The calculated heats of formation of Sb–Y system with experimental data at 300 K [16].

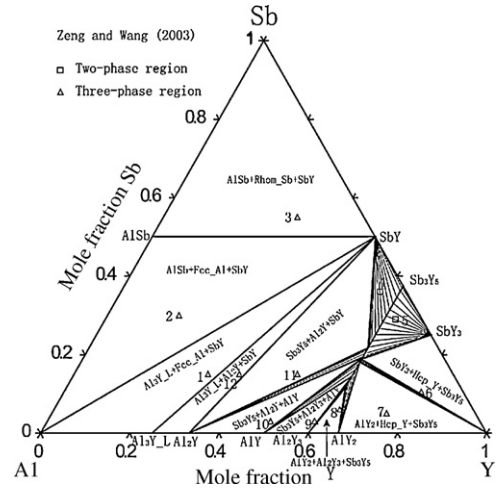

Fig. 5. The calculated isotherm section of Al–Sb–Y system at 800 K with experimental data [18].

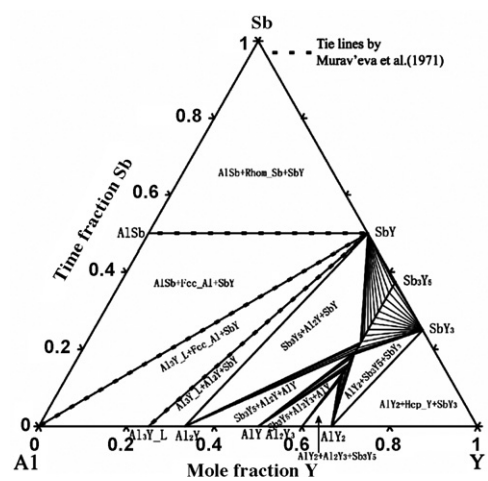

Fig. 6. The calculated isotherm section of Al–Sb–Y system at 773 K.

as show in Figs. 7 and 8. And the calculated invariant reactions and temperatures involved liquid in the Al–Sb–Y ternary system are summarized in Table 5. Further experimental data are needed to verify the liquids projection.

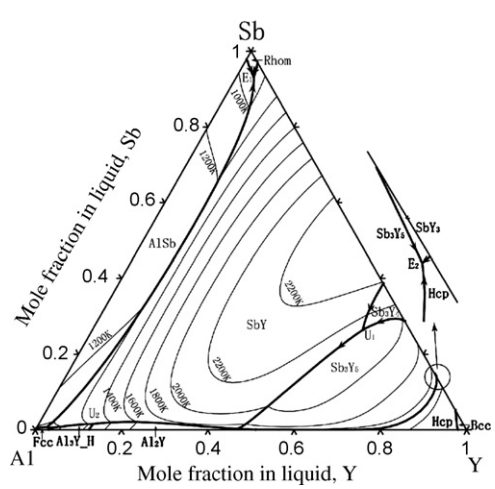

Fig. 7. The calculated liquidus projection of Al–Sb–Y system.

Table 4
Comparison of calculated and experimental results.

Phase regions	Alloy composition (at.%)			Experimental results	Calculated results
	Al	Sb	Y		
1	55	15	30	$Al + Al_3Y + SbY$	$Al + Al_3Y_L + SbY$
2	54	30	16	$Al + AlSb + SbY$	$Al + AlSb + SbY$
3	15	55	30	$Sb + AlSb + SbY$	$Sb + AlSb + SbY$
4	6	36	58	$SbY + Sb_3Y_5$	$SbY + Sb_3Y_5$
5	6	29	65	$Sb_3Y_5 + SbY_3$	$Sb_3Y_5 + SbY_3$
6	10	10	80	$Y + Sb_3Y_5 + SbY_3$	$Y + Sb_3Y_5 + SbY_3$
7	20	5	75	$\mathit{AlY_3} + Y + Sb_3Y_5$	$\mathit{AlY_2} + Y + Sb_3Y_5$
8	30	5	65	$Al_2Y_3 + \mathit{AlY_3} + Sb_3Y_5$	$Al_2Y_3 + \mathit{AlY_2} + Sb_3Y_5$
9	37	3	60	$AlY + Al_2Y_3 + Sb_3Y_5$	$AlY + Al_2Y_3 + Sb_3Y_5$
10	47	3	50	$AlY + Al_2Y + Sb_3Y_5$	$AlY + Al_2Y + Sb_3Y_5$
11	35	15	50	$SbY + Al_2Y + Sb_3Y_5$	$SbY + Al_2Y + Sb_3Y_5$
12	48	15	37	$SbY + Al_2Y + Al_3Y$	$SbY + Al_2Y + Al_3Y_L$

The italicized means the difference between the calculated result and experimental data.

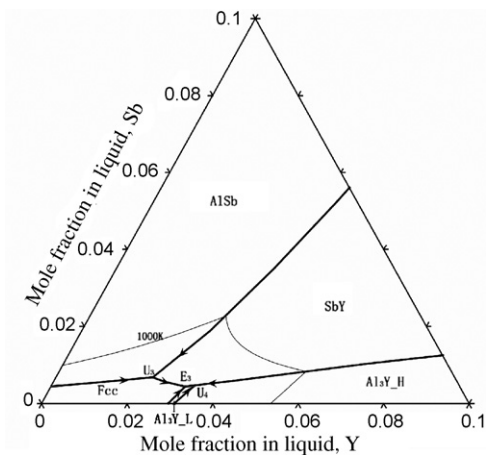

Fig. 8. The calculated liquidus projection of Al-rich region.

Table 5
Calculated invariant reactions and temperature of Al–Sb–Y ternary system.

Type	Reaction	T (K)
E_1	Liquid ⇔ Rhom_Sb + AlSb + SbY	887.10
E_2	Liquid ⇔ Hcp_Y + Sb_3Y_5 + SbY_3	1498.24
E_3	Liquid ⇔ Fcc_Al + Al_3Y_L + SbY	908.06
E_4	Liquid ⇔ AlY + Al_2Y_3 + Sb_3Y_5	1356.30
U_1	Liquid + Sb_3Y_4 ⇔ SbY + Sb_3Y_5	2058.73
U_2	Liquid + Al_2Y ⇔ Al_3Y_H + SbY	1233.14
U_3	Liquid + AlSb ⇔ Fcc_Al + SbY	915.54
U_4	Liquid + Al_3Y_H ⇔ Al_3Y_L + SbY	915.58
U_5	Liquid + SbY ⇔ Al_2Y + Sb_3Y_5	1511.33
U_6	Liquid + Al_2Y ⇔ Sb_3Y_5 + AlY	1409.63
U_7	Liquid + Al_2Y_3 ⇔ Sb_3Y_5 + AlY_2	1253.86
U_8	Liquid + Sb_3Y_5 ⇔ AlY_2 + Hcp_Y	1229.63

5. Conclusions

The Al–Sb–Y ternary system had been assessed thermodynamically based on reported experimental data of phase diagrams and thermodynamic properties. Reasonable agreement between calculated and experimental data had been reached and thermodynamic parameters for various phases in this ternary system had been obtained.

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